TEXORA · LECTURE SHEET

Equipment Used in Preparatory Wet Processes

The machines that carry out preparation: how the form of the material (yarn, woven, knit) and the running mode (batch or continuous) choose the machine, and the construction, schematic arrangement, speed and capacity of the padder, the J-box and the kier.

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1. Two ways of organising the work

- **Batch-wise processing:** a fixed load of material is treated with a bath in one vessel through the whole step (winch, jigger, kier, package machine). Flexible for small lots and shade changes, but liquor ratios are high and handling is heavy.
- **Continuous processing:** fabric in open width runs steadily through a train — padder → steamer/J-box → washing compartments — at tens of metres per minute. Uniform and economical for long runs of one fabric; changeovers are slow.
- **Semi-continuous** variants (pad-batch: pad, batch on a rotating roll for hours, then wash) sit between the two.

The form of material decides the vessel: yarn is processed as hanks (in cabinets) or wound on perforated packages (liquor pumped through the package); woven fabric runs in **rope form** (winch) or **open width** (jigger, continuous ranges); knits, which crease and stretch easily, are handled in rope form in low-tension winches and soft-flow machines.

2. The padder (padding mangle)

Function: to impregnate fabric uniformly with a chemical liquor and squeeze out the excess to a controlled, reproducible amount. Every continuous preparation step (desize, scour, bleach) and pad-dyeing/ pad-batch process begins here.

Construction: a stainless-steel trough holds the liquor; guide rollers lead the fabric down into the trough around an immersion roller so it is fully wet, then vertically up through the **nip** between two (or three) rubber-covered **bowls**. The bowls are pressed together by pneumatic/hydraulic loading (and often crowned or deflection-controlled) to give an even nip pressure across the width. Liquor expressed at the nip falls back into the trough. Typical running speeds 20–120 m/min according to the range it feeds.

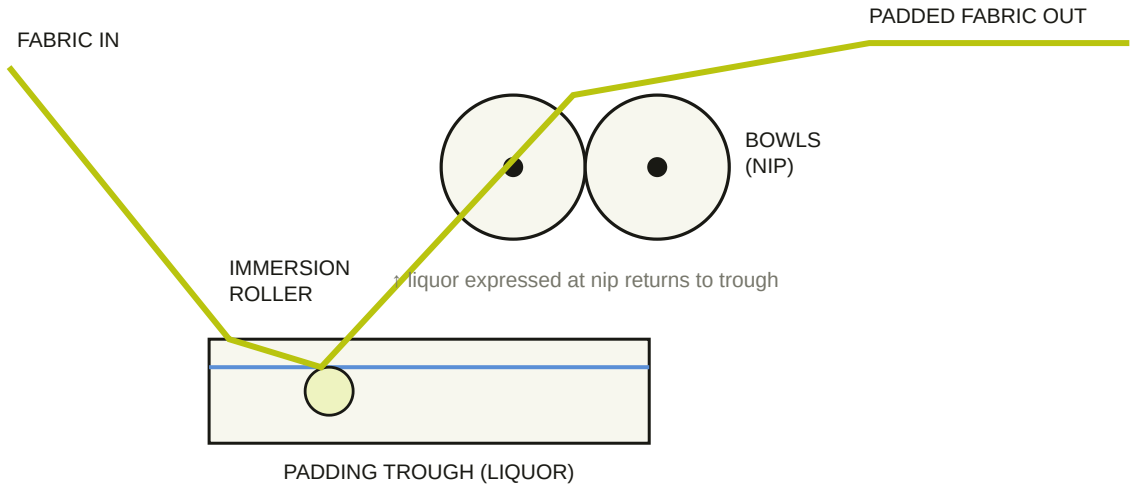


Fig. 5 — Two-bowl padding mangle: trough, immersion roller and nip (schematic).

The controlled result is the **pick-up** (expression):

$$\text{Pick-up \%} = (\text{wet weight after padding} - \text{dry weight}) \div \text{dry weight} \times 100$$

WORKED EXAMPLE

100 kg of dry fabric weighs 175 kg after padding with scouring liquor containing 40 g/L NaOH. Pick-up = (175 − 100)/100 × 100 = **75%**. The fabric carries ≈ 75 L of liquor, hence 75 × 40 = 3000 g = **3 kg NaOH, i.e. 3% on weight of fabric** — the figure the recipe is actually built on. Low, uneven pick-up means low, uneven chemical add-on, so nip uniformity is a preparation quality in itself.

3. The kier

Function: the classic batch vessel for scouring (kier boiling) and bleaching of cotton yarn and fabric, working at or above the boil under pressure. **Construction:** a vertical (or horizontal) cylindrical steel vessel with a dished lid, a perforated **false bottom** on which the goods rest, and a liquor circulation system: liquor is drawn from below the false bottom, pumped through a heater (steam coil or external exchanger) and sprayed over the top of the load, percolating down through the goods continuously. Vents release air (oxygen in the liquor damages cellulose at the boil — modern kiers are filled and purged to avoid it). Capacity typically **500–2000 kg** per batch; a cotton scouring cycle runs 4–8 h at 100–130 °C plus loading and washing time.

Fig. 6 — Vertical kier with external liquor circulation and heating (schematic)

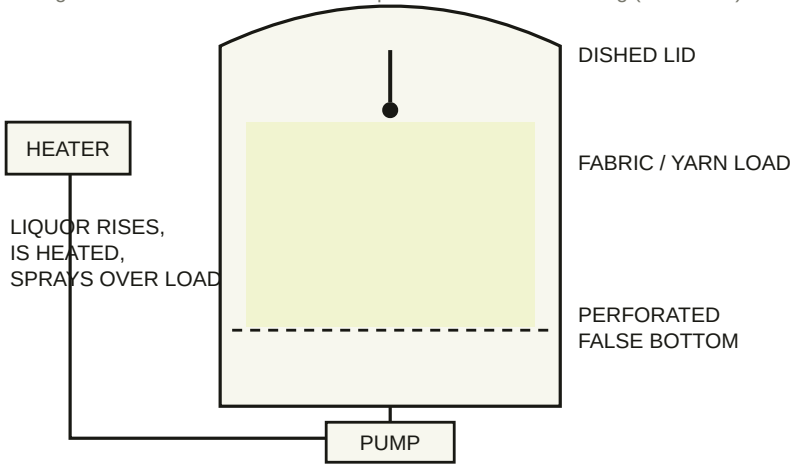


Fig. 6 — The kier: goods rest on a false bottom while liquor circulates, heats and showers over them.

4. The J-box

Function: the reaction chamber of continuous preparation. Fabric leaving the padder is plaited (folded in loose layers) into the top of a tall J-shaped stainless chamber, travels slowly down the vertical leg, around the J-bend and up the shorter leg while the chemical reaction (scouring or bleaching) proceeds — in steam-heated versions at ≈ 100 °C. Because the chamber holds a large mass of fabric, residence times of 20–60 min (pad-steam) up to 1–3 h (storage types) are available at production speeds the padder alone could never allow.

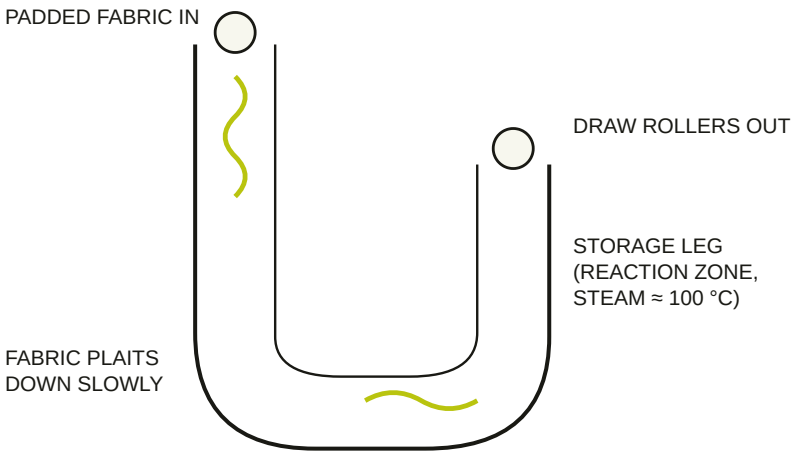


Fig. 7 — J-box: long dwell in a compact chamber (schematic)

Fig. 7 — The J-box gives minutes-to-hours of reaction time in a continuous line.

WORKED EXAMPLE

A J-box holds 900 kg of fabric and the range runs at 30 kg/min. Dwell time = 900 ÷ 30 = **30 minutes** — enough for a pad-steam scour. To double the dwell at the same speed, the box must

hold 1800 kg; to keep dwell with twice the throughput, capacity must double. Capacity, speed and dwell are one equation: dwell = hold-up ÷ throughput.

5. Machines for yarn and knit preparation

- **Hank (skein) machines:** hanks hang on sticks or poles in a cabinet; liquor circulates around them. Gentle; used for yarn scouring/bleaching at small scale.
- **Package machines:** yarn wound on perforated cones/cheeses stacked on spindles in a sealed vessel; liquor is pumped inside-out and outside-in through the packages. The workhorse for yarn preparation and dyeing — uniform if package density is even.
- **Winch:** fabric in rope form circulates through a bath over a driven reel (elliptical winch reel pulls gently). High liquor ratio (1 : 15–1 : 30), capacities ≈ 100–500 kg; traditional for knits and delicate wovens.
- **Soft-flow / jet machines:** rope fabric is transported by the liquor stream itself — much lower tension, shorter baths; the modern standard for knit preparation and dyeing.
- **Jigger:** open-width fabric winds back and forth between two rollers through a small trough of liquor — low liquor ratio and full width control; capacities of a few hundred metres to over a thousand metres per batch; used for woven preparation steps and dyeing where creasing must be avoided.

6. Machine comparison (typical guide values)

MACHINE	FORM	MODE	TYPICAL SPEED / CYCLE	TYPICAL CAPACITY
Padder	Open width	Continuous	20–120 m/min	Sets the range speed; pick-up 60–100%
J-box	Open width	Continuous	Dwell 20 min – 3 h	Hold-up ≈ 500–2000 kg
Kier	Yarn / fabric load	Batch	4–8 h boil + handling	500–2000 kg per batch
Winch	Rope	Batch	Bath cycles 30 min – hours	100–500 kg; liquor 1 : 15–1 : 30
Jigger	Open width	Batch	Passes of minutes each way	Few hundred to > 1000 m per batch
Package machine	Yarn packages	Batch	Cycles 1–4 h	100–1000 kg

7. Glossary — key terms

- **Pick-up / expression:** liquor carried by fabric after padding, as % of dry weight.
- **Rope form / open width:** fabric gathered lengthwise into a rope, versus held flat at full width.
- **False bottom:** perforated plate in a kier supporting the load while liquor drains through.
- **Dwell (residence) time:** how long material stays inside a reaction chamber.
- **Pad-batch:** pad, then store the roll (often rotating) for hours before washing off.

8. Key points

- Batch vessels (winch, jigger, kier, package) trade flexibility for high liquor use; continuous ranges (padder + J-box/steamer + washers) trade changeover speed for uniformity and economy.
- The padder controls chemical add-on through pick-up — recipes are calculated from it.
- The kier cooks a static load with circulating, heated liquor; the J-box stores moving fabric long enough for the reaction.
- Machine choice follows material form: packages for yarn, open width for wovens, low-tension rope machines for knits.

9. Summary

Preparation equipment is organised around a simple logic. Liquor must be applied uniformly (the padder, whose pick-up fixes the dose), given time and temperature to react (the kier for batch loads, the J-box for continuous flow), and washed out efficiently (wash ranges, winches, jiggers). Capacities, speeds and liquor ratios in the comparison table are the numbers that decide which machine a given fabric, lot size and preparation recipe belongs on.

10. Practice

Answer in your own words; brief model answers follow.

1. A padder delivers 65% pick-up of a bleach liquor containing 30 mL/L of hydrogen peroxide solution. How much peroxide solution does 200 kg of fabric carry into the steamer?
2. Why does a knit fabric go to a soft-flow machine rather than a jigger for preparation?
3. In a kier, what two jobs does the liquor-circulation system do, and why is air excluded from the vessel during boiling?
4. A continuous bleach range must give 45 min dwell at 25 kg/min. What fabric hold-up must the J-box provide?

5. State one advantage and one disadvantage of continuous preparation against batch preparation.

Answer key

1. Picked-up liquor = 200 × 0.65 = 130 L. Peroxide solution carried = 130 × 30 mL/L = 3900 mL = **3.9 L**.
2. Knits stretch and crease under tension and show permanent crease marks; the jigger holds fabric flat under winding tension, while the soft-flow transports the rope in a liquor stream at very low tension, protecting structure and surface.
3. It heats the load evenly (through the heater) and keeps fresh liquor percolating through every part of the static load so treatment is uniform. Air is excluded because dissolved oxygen in hot alkali oxidises cellulose (oxycellulose tendering), especially in the presence of metal ions.
4. Hold-up = dwell × throughput = 45 × 25 = **1125 kg**.
5. Advantage (any): uniform treatment and low unit cost/liquor use on long runs. Disadvantage (any): slow, wasteful changeovers — unsuited to small lots and frequent fabric changes; a fault can affect a long length before it is seen.

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